MIL-T-27179B (USAF)
22 December 1969
Superseding
MIL-T-27179A (USAF)
20 September 1961

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, TRU-27/A, ENGINE

1. SCOPE

1.1 This specification covers the requirements for a turbo-jet engine, aircraft rate-of-flow transmitter, designated TRU-27/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS
Federal

L-P-378	Plastic Film (Polyethylene Thin Gauge)
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

SPECIFICATIONS
Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-J-5624	Jet Fuel, Grades JP-3, JP-4, and JP-5
MIL-I-7566	Indicator System, Fuel Flowmeter, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-F-25656	Fuel, Aircraft Turbine and Jet Engine, Grade JP-6
MIL-C-26500	Connectors, General Purpose, Electrical, Miniature Circular, Environment Resisting, 200 Deg. Ambient Temperature
MIL-M-45202	Magnesium Alloy, Anodic Treatment of

FSC 6620

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STANDARDS
Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MS 33649	**Bosses,** Fluid Connection, Internal Straight Thread

*(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids.

3.2 General specification. The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and Construction.

3.3.1 Fuel variation. Unless otherwise specified, the transmitter shall be designed to operate satisfactorily with fuels having any of the variations in characteristics specified in MIL-J-5624 and MIL-F-25656. The transmitter shall be calibrated for rate of flow in pounds per hour and shall measure the fuel flow rate on a gravimetric basis without separate and distinct density correcting means.

*3.3.2 Indicators. The transmitter shall be designed to operate within the tolerances specified herein and with fuel flow indicators having a range of between 400 and 20,000 pph. The transmitter shall provide a 5-wire synchro output for the indicators which shall have electrical characteristics conforming to Clifton Precision Products Company's type TGC-11-E-6 synchro torque transmitter, or equal. The transmitter shall be so calibrated that the rotor of the output synchro transmitter will rotate over a range of 128° for a corresponding fuel flow range of 0 to 16,000 pph sensed by the transmitter.

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3.3.2.1 When two indicators are operated from the one transmitter output, the resulting load errors shall not exceed 1° of arc over the errors resulting when only one indicator is used.

3.3.3 Shaft rotation. For an increasing flow rate, the direction of rotation of the transmitter synchro shaft shall be clockwise as viewed from the shaft end.

*3.3.4 Power. The transmitter shall be designed to operate from 115V 400Hz, ±5 percent, single-phase a-c aircraft power and as otherwise stated in MIL-STD-704. The wave form may be nonsinusoidal with a crest factor of 1.414 ±10 percent and with harmonics with respective amplitudes of less than 5 percent. No damage to the transmitter shall result if a voltage and frequency of 10 percent more or less than the tolerances specified is applied.

3.3.5 Pressures. The transmitter shall be designed for operation in fuel systems with a normal fuel pressure up to 600 psi and shall withstand high pressure of 1,600 psi at room temperature and 800 psi at -54° C without operational damage.

3.3.6 Electrical receptacles. The transmitter shall be provided with a hermetically sealed, miniature, electrical receptacle in accordance with MIL-C-26500 and shall be designed to meet the environmental requirements of this specification.

*3.3.7 Internal wiring. Where practicable, the internal wiring shall be color coded. The receptacle shall be marked as follows:

PIN	CONNECTION
1 or A	Power ground side
2 or B	Power input 115V 400 Hz to transmitter motor and synchro rotor
3 or C	X_2 of transmitter synchro stator
4 or D	Z_2 of transmitter synchro stator
5 or E	Y_2 of transmitter synchro stator
6 or F	Spare
7 -	Spare (optional).

*3.3.8 Electrical zero. The synchros in the transmitter and fuel flow indicator shall be so oriented that their electrical zero will correspond to a fuel flow of 10,000 pph. Identification of the synchro leads and the determination of electrical zero shall be as follows:

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a. The rotor leads shall be designated as H_2 and C_2, and the C_2 lead shall be grounded.

b. One stator lead shall be designated as Z_2 and connected to the C_2 rotor lead.

c. Rated voltage shall be applied across H_2 and C_2 and the voltage measured across H_2 and one of the unlabeled stator leads and across the two unlabeled stator leads.

d. The rotor shall be rotated until the voltage measured across H_2 and the one unlabeled stator lead is approximately a minimum and the voltage across the two unlabeled stator leads is exactly minimum. This position is the electrical zero position corresponding to 10,000 pph fuel flow.

e. The rotor shall be rotated in a clockwise direction from electrical zero, as viewed from the rotor shaft end. The voltage across one of the unlabeled stator leads and Z_2 shall increase and the voltage across the other unlabeled stator lead and Z_2 shall decrease. The stator lead for increasing voltage shall be labeled X_2 and the stator lead for decreasing voltage shall be labeled Y_2.

3.3.9 Pressure drop. The transmitter shall be so designed that when inoperative, under any fuel flow conditions, the pressure will not be appreciably more than when the transmitter is operating normally.

3.4 Performance.

3.4.1 Power consumption. Power consumption shall not exceed 40 va.

3.4.2 Range. The transmitter shall measure mass flow rates from 500 to 20,000 pph.

3.4.3 Temperature range. The transmitter shall operate satisfactorily in ambients from -54° to +121° C and shall withstand exposure to a temperature of 190° C. Fuel ambients through the transmitter shall range from -54° to +71° C.

*3.4.4 Tolerances. When subjected to the tests specified in MIL-I-7566 and herein, the tolerances shall be within the limits specified in tables I through IV.

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TABLE I

FUEL FLOW TRANSMITTER SCALE ERROR TOLERANCES

Fuel Flow PPH	Tolerance ±PPH	
	Room Temp	High, Low, and Substitute Fuel
0-500	120	150
1,000	50	70
2,000	25	40
3,000	30	50
4,000	30	50
6,000	50	70
9,000	80	100
12,000	110	150
15,000	140	180
20,000	190	240

TABLE II

TRANSMITTER ACCURACY

Test	Tolerance ±PPH
Voltage and frequency error	5 percent of point
Position error	25
Friction	
Normal, room, high, low	20
Vibration	
Oscillation	50
Variation	50

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TABLE III

FLOW RATE AND PRESSURE DROP

Flow Rate PPH	Maximum Allowable Pressure Drop PSI
10,000 20,000	0.25 1.0

TABLE IV

CONTAMINATED FUEL PARTICLE SIZE

Foreign Particle Size, Microns	Percent of Total Weight
0-5	39 ±2
5-10	18 ±3
10-20	16 ±3
20-20	18 ±3
over 40	9 ±3
Through a 200 mesh screen	100 percent

*3.4.5 Flame. The transmitter shall withstand a 1093° C flame for 5 minutes.

3.5 Mechanical stop. A suitable mechanical stop shall be provided within the transmitter above the 20,000 pph point. The transmitter shall not be damaged by fuel flow rates exceeding 20 percent of the rated value, provided the pressure rating of the transmitter is not exceeded.

3.6 Dimensions. The outline dimensions of the transmitter shall conform essentially to figure 1.

*3.7 Weight. The weight of the transmitter shall not exceed 7.75 pounds.

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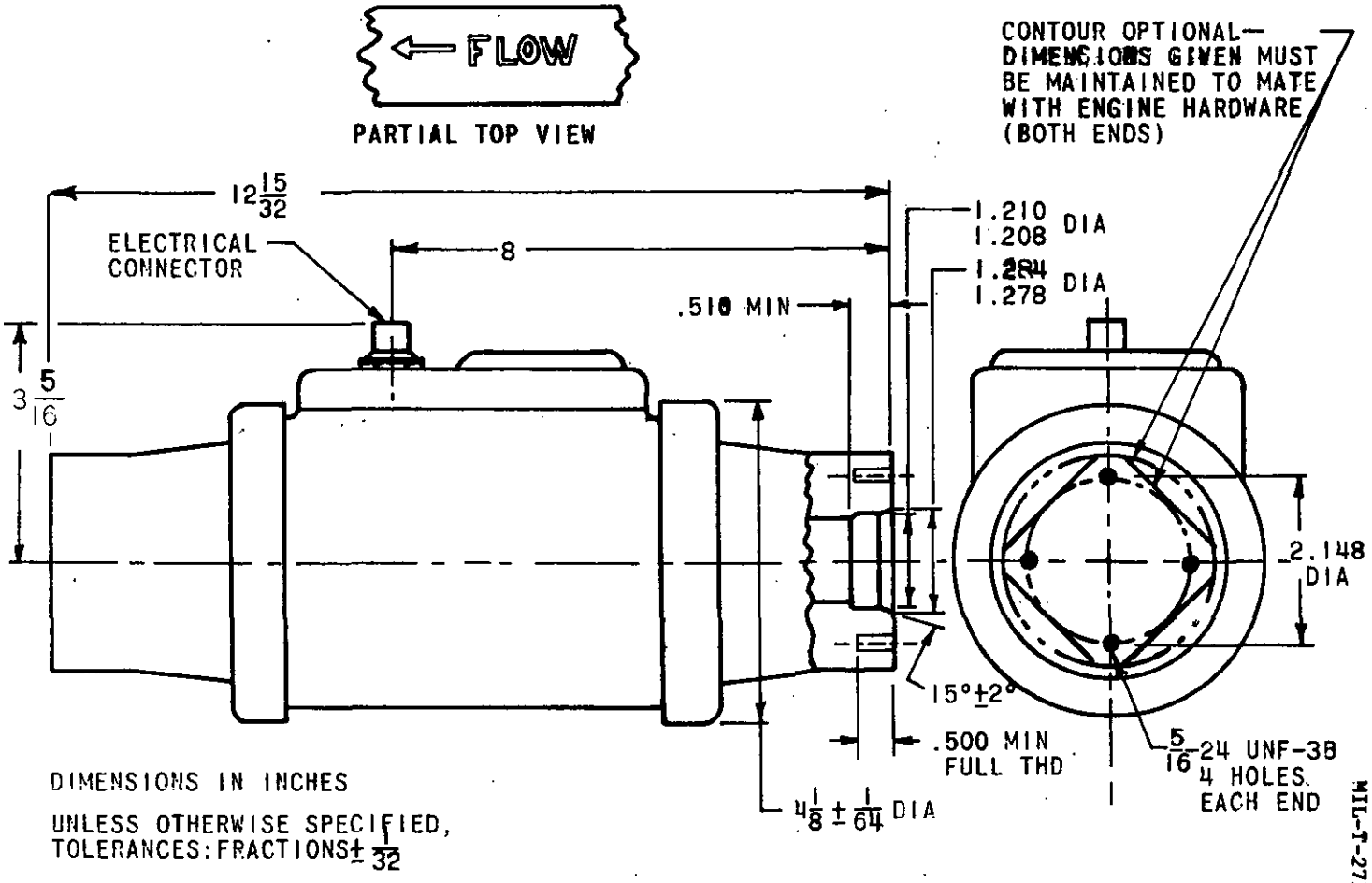


FIGURE 1 OUTLINE DIMENSIONS, FLOWMETER TRANSMITTER

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3.8 Flow direction. The direction of flow through the transmitter shall be permanently and legibly marked on both front and back of the transmitter by an arrow as shown in figure 1.

3.9 Caps. The fuel line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

*3.10 Mounting. The transmitter fitting shall be designed for connection to a 1-1/4-inch diameter fuel line. End bosses shall be in accordance with MS33649.

3.11 Material. Where high temperature and other stringent environmental conditions are a factor in the design of the transmitter, the use of titanium material for lightweight and high-strength characteristics will be permitted. The use of lightweight materials and weight-saving designs shall be a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.12 Finishes and Protective Coatings.

3.12.1 Alloy parts. Where practicable, aluminum or titanium alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

3.12.2 Chemical treatment. A chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for those alloys which do not anodize satisfactorily where the primary purpose of the treatment is to afford a suitable print base, or for castings containing nonaluminum or titanium alloy inserts.

3.12.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

*3.12.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202.

3.13 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

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*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General specification. The inspection and testing of transmitters shall be in accordance with MIL-I-7566 and as specified herein. The applicable tolerances listed in Tables I through IV shall apply.

4.3 Test Conditions.

4.3.1 Test power. The supply power shall be single-phase ac, 115V ±2 percent, and 400 Hz, controlled to less than ±1 Hz, and of known value.

4.3.2 Test fuel. Unless otherwise specified, the fuel used in testing for calibration purposes shall be JP-4 conforming to MIL-J-5624. The specific gravity of the fuel shall be 0.785 ± 0.005 at 15.6° C. (Note: A commercial equivalent with higher flash point may be used.)

4.3.3 Test position. Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.3.4 Test temperature. Unless otherwise specified, all tests shall be conducted at room temperature.

4.4 Test Methods.

4.4.1 Sampling Plans.

4.4.1.1 Sampling Plan A.

4.4.1.1.1 Position error. Position error shall be determined at the 4,000 pph flow rate. Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the value specified in Table II.

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4.4.1.1.2 Response time. The time required for the transmitter output signal to come within 1 percent of the final steady state signal shall not exceed 2 seconds with a step change in flow rate from 500 to 20,000 pph.

4.4.1.1.3 Vibration. Vibration testing shall be conducted in accordance with MIL-E-5272, procedure XII, curve A, at room temperature, except the vibration schedule shall be as follows. The fuel flow through the transmitter shall be approximately 4,000 pph.

0.036 da	5 to 52 Hz
±5g	52 to 130 Hz
±10g	130 to 185 Hz
±12g	185 to 2,000 Hz

Pointer oscillation and variation shall not exceed the tolerances specified in Table II, except that a maximum of two resonances will be permitted with an oscillation or variation tolerance of twice that specified in Table II, if oscillation and variation of this magnitude are not obtained simultaneously. At the completion of this test, the transmitter shall be subjected to the scale error and friction tests. The errors shall not exceed those specified in Tables I and II.

4.4.1.1.4 Leak test. Testing shall be as specified in MIL-I-7566 for high-pressure type transmitters.

4.4.1.1.5 Scale error. Scale error tests shall be conducted at room, low, and high temperatures and substitute fuel operating conditions. Tolerances shall not exceed those specified in Table I.

4.4.1.1.6 Friction error. Friction error shall be determined at the 4,000 and 8,000 pph flow rates. The differences in readings taken before tapping and after tapping shall not exceed the values listed in Table II. This test may be combined with the position error test specified in 4.4.1.1.1.

4.4.1.1.7 Voltage and frequency variation effect. The applied voltage and frequency for this test shall be set at the following points.

109V ac 380 Hz	109V ac 420 Hz
121V ac 380 Hz	121V ac 420 Hz

The voltage and frequency test shall be conducted at the 5, 50, and 90 percent of full scale flow rates. Readings shall be taken at the 115V 400 Hz normal condition and at the ±5 percent conditions tabulated herein.

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The reading taken at each of the above conditions shall not differ from the reading taken at the normal condition in excess of the value shown in Table II. The test shall be conducted with a master indicator in accordance with MIL-I-7566.

4.4.1.2 Sampling Plan B.

*4.4.1.2.1 High temperature exposure. The transmitter, while inoperative, shall be subjected to a temperature of 121° ±2° C for a period of 4 hours. The temperature shall then be raised to 190° ±2° C for a period of 30 minutes. This cycle shall be repeated 6 times for a total of 24 hours at 121° C and 3 hours at 190° C. The transmitter shall then be energized and immediately tested for scale error and friction at 70° ±2° C air ambient and 38° ±2° C fuel. The tolerances shall not exceed those specified in Tables I and II. The transmitter shall be examined for evidence of failure. During temperature cycling, fluid may be inside the transmitter but not flowing.

4.4.1.2.2 Contaminated fuel operation. Test fuel containing dirt to the extent of 80 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. A scale and friction error test shall then be conducted using the contaminated fuel. Operation shall be satisfactory and within the room temperature tolerances. This foreign matter shall consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as shown in Table IV.

4.4.1.2.3 Shock. The shock test shall be in accordance with procedure IV of MIL-E-5272. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration 7 ±2 milliseconds. Three shocks shall be applied in each of the following directions:

a. Along the longitudinal axis.

b. At 45° to the longitudinal axis in one direction.

4.4.1.2.4 Substitute fuel. The substitute fuel test shall be conducted using JP-6 fuel as specified in MIL-F-25656. This test shall be conducted under room operating temperature.

4.4.1.2.5 Altitude. No altitude test shall be required with fuel flow. Electrical components that are not hermetically sealed shall be subjected to an 80,000-foot altitude test.

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*4.4.1.2.6 Flame test. While inoperative, but with a fluid flow of 6,000 pph at 50 psig, the transmitter shall be subjected to a 1093° C flame for a period of 5 minutes. Any physical damage that would result in leakage sufficient to sustain a flame shall be cause for rejection.

*4.5 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Each transmitter to be preserved and packaged in quantity unit packs of one each in accordance with Method 1C-2 of MIL-P-116. Transmitter to be wrapped with L-P-378 and blocked in position using cells, sleeves or die cuts constructed in accordance with PPP-F-320 within the first container required by the method of preservation. Completed packed to be overboxed within a PPP-B-636 container for added protection. The above packaging requirements are mandatory unless otherwise stipulated by the procuring activity.

*5.2 Packing. Each transmitter shall be packed Level B for Air Force stock or shipment as defined in MIL-STD-794 unless otherwise specified in the contract or order by the procuring activity.

*5.3 Marking of shipments. Marking of shipments shall be in accordance with MIL-I-7566. The nomenclature shall be as follows:

Transmitter, Rate of Flow TRU-27/A, Engine

6. NOTES

6.1 Intended use. The TRU-27/A transmitter covered by this specification is intended to measure and transmit a signal proportional to the total mass rate of fuel flow to the main engine of jet aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this document.

b. Whether Sampling Plan B tests are to be conducted.

c. Selection of applicable levels of packaging and packing.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OCAMA (OCNE), Tinker Air Force Base, Oklahoma 73145, and information pertaining to qualification of products may be obtained from that activity.

Note: The margins of this specification are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 71

Preparing Activity:
Air Force - 71

Reviewer:
Air Force - 71

Project Number:
6620-F039

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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